ITA06020 MACHINE LEARNING

Experiment 9 – Comparison of Linear and Polynomial Regression

Aim:

To implement and compare Linear Regression and Polynomial Regression algorithms in Python using a real-world dataset.

Dataset: Diabetes Dataset (available in Scikit-learn).

We will use BMI (Body Mass Index) to predict disease progression.

Procedure

1. Import the required Python libraries.
2. Load the Diabetes dataset.
3. Select BMI as the input feature and disease progression as the target.
4. Split the dataset into training and testing data.
5. Apply Linear Regression.
6. Apply Polynomial Regression with degree 2.
7. Predict the outputs using both models.
8. Compare their performance using Mean Squared Error and R^2 score.

Python Program

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.pipeline import make_pipeline

from sklearn.metrics import mean_squared_error, r2_score

data = load_diabetes()

X = data.data[:, [2]]

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.2, random_state=42

)

linear = LinearRegression()

linear.fit(X_train, y_train)
```

```
linear_pred = linear.predict(X_test)
poly = make_pipeline(
   PolynomialFeatures(degree=2),
   LinearRegression()
)
poly.fit(X_train, y_train)
poly_pred = poly.predict(X_test)
print("Linear Regression")
print("Actual Output:", y_test[0])
print("Predicted Output:", round(linear_pred[0], 2))
print("MSE:", round(mean_squared_error(y_test, linear_pred), 2))
print("R2 Score:", round(r2_score(y_test, linear_pred), 4))
print("\nPolynomial Regression")
print("Actual Output:", y_test[0])
print("Predicted Output:", round(poly_pred[0], 2))
print("MSE:", round(mean_squared_error(y_test, poly_pred), 2))
print("R2 Score:", round(r2_score(y_test, poly_pred), 4))
```

Output:

```
Linear Regression
Actual Output: 219.0
Predicted Output: 145.81
MSE: 4061.83
R2 Score: 0.2334

Polynomial Regression
Actual Output: 219.0
Predicted Output: 144.32
MSE: 4085.03
R2 Score: 0.229
```

Result

Thus, Linear Regression and Polynomial Regression were successfully implemented and compared using the Diabetes dataset. Their performance was evaluated using Mean Squared Error and R^2 score.